ECU Diagnostics & Validation Reference (Sample)

Synthetic educational document for the DiagPilot RAG corpus. Not Bosch documentation and contains no proprietary data. Always defer to the project's controlled specifications.

1. DTC Structure and Status

A diagnostic trouble code (e.g., P0335) encodes a system letter (P powertrain, C chassis, B body, U network), a standard/manufacturer-specific digit, a subsystem digit and a fault index. Status moves from **pending** (seen once) to **confirmed** (seen across enough driving cycles or debounce counts) and may set a warning lamp. Freeze-frame data stores a few signals at the moment of the fault and is a snapshot, not a trace.

2. Common UDS Services (ISO 14229)

Service	ID	Use
DiagnosticSessionControl	0x10	Switch session
ReadDTCInformation	0x19	Read DTCs and status
ReadDataByIdentifier	0x22	Read a data value
ClearDiagnosticInformation	0x14	Clear DTCs (loses evidence)
RoutineControl	0x31	Run a test routine

Clearing DTCs before logging them destroys evidence and should be avoided during an open investigation.

3. Triage Method for a Position-Sensor DTC (e.g., P0335, P0340)

Check in order: (1) power supply and ground quality (cranking voltage sag), (2) wiring and connector, (3) sensor air gap and target wheel, (4) signal amplitude against threshold, (5) software debounce and threshold changes between releases. Compare a failing run against a passing baseline with the same ECU variant and conditions before concluding a root cause.

4. Regression Analysis Between Software Releases

When a test passes on one release and fails on the next, list the release-note changes touching the affected function first. Correlation is not causation: confirm by controlled A/B on the same hardware, ambient condition and battery state.

5. Functional Safety Note (ISO 26262)

Safety classification (ASIL), safety-goal impact, and release sign-off are decisions owned by the responsible safety engineer and the project's safety case. A diagnostic assistant may summarise evidence but must not approve, classify or sign off.

6. Calibration and Flashing

Changes to calibration parameters or flashing software to a test vehicle follow the project change-control process, with an approved ticket, and are never executed automatically.

7. Reporting

A good defect report states ECU variant, SW version, test conditions, DTCs with timestamps, relevant signal evidence, comparison to baseline, hypothesis with confidence, and what is not yet known.